

20-Day Complete Machine Learning Revision Planner

Mathematics + Core ML + Advanced ML + Projects + Interview Preparation

Daily Target: 4–5 hours

60 min Mathematics **90 min** ML Theory **90 min** Coding **30 min** Interview Questions **30 min** Active Recall

How to Use This Planner

For every algorithm, follow:

1. **What?** – Define the algorithm in your own words.
2. **Why?** – Know when and why it is useful.
3. **How?** – Understand the working and intuition.
4. **Code** – Implement it with Python/scikit-learn; implement key concepts from scratch.
5. **Explain** – Close your notes and teach the concept aloud.

Revision rule: Learn → Close Notes → Recall → Code → Debug → Explain → Interview Question.

DAY 1 – Linear Algebra for ML**Phase: Mathematical Foundation****Mathematics**

- Scalars, vectors and matrices
- Vector addition and scalar multiplication
- Dot product and matrix multiplication
- Transpose, identity matrix and inverse
- Rank and linear independence
- L1 and L2 norms

Key formulas

$$a \cdot b = \sum_i a_i b_i, \quad \|x\|_2 = \sqrt{\sum_i x_i^2}$$

ML Connection

- Dataset as a matrix
- Samples as rows and features as columns
- Model parameters as vectors
- Distance and similarity

Coding

- NumPy dot product
- Matrix multiplication
- Transpose
- Euclidean distance
- Cosine similarity

Interview

- Why is linear algebra important in ML?
- Dot product vs matrix multiplication
- What is a vector norm?
- What does matrix rank mean?

Daily Output: One-page linear algebra cheat sheet + NumPy practice.

DAY 2 – Calculus + Optimization**Mathematics**

- Functions and derivatives
- Partial derivatives
- Gradients
- Chain rule
- Optimization
- Gradient descent
- Learning rate

Key formula

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla J(\theta)$$

Understand

- Loss $\rightarrow$ derivative $\rightarrow$ gradient $\rightarrow$ parameter update
- Batch, stochastic and mini-batch gradient descent

Coding

- Implement linear regression from scratch
- Implement gradient descent without sklearn
- Plot loss versus iterations

Interview

- What is a gradient?
- What if learning rate is too large or too small?
- Batch GD vs SGD vs mini-batch GD

Daily Output: Working gradient-descent implementation.

DAY 3 – Probability + Statistics**Probability**

- Probability and conditional probability
- Independence
- Random variables
- Bayes theorem
- Expected value
- Variance and covariance

Key formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Statistics

- Mean, median and mode
- Variance and standard deviation
- Percentiles
- Normal distribution
- Z-score
- Correlation

ML Connection

- Naive Bayes
- Feature correlation
- Standardization
- Probability predictions

Coding: Calculate statistical quantities using NumPy/Pandas on a real dataset.

Daily Output: Probability + statistics formula sheet.

DAY 4 – ML Fundamentals + Data Preprocessing**Theory**

- AI vs ML vs DL
- Supervised, unsupervised and reinforcement learning
- Features, labels, training, validation and testing
- Generalization

Preprocessing

- Missing values and duplicates
- Outliers
- Categorical encoding
- Scaling
- Feature engineering
- Feature selection
- Normalization vs standardization

Key formula

$$z = \frac{x - \mu}{\sigma}$$

Coding

- Build preprocessing with `Pipeline`
- Use `ColumnTransformer`
- Prepare a complete train/test pipeline

Daily Output: Reusable preprocessing pipeline.

DAY 5 – Linear Regression + Regularization**Topics**

- Simple and multiple linear regression
- Polynomial regression
- Cost function
- Gradient descent
- Normal equation
- Regression assumptions

Regularization

- Ridge regression (L2)
- Lasso regression (L1)
- Bias–variance intuition

$$\text{Ridge: Loss} + \lambda \sum_i w_i^2 \quad \text{Lasso: Loss} + \lambda \sum_i |w_i|$$

Coding

- Linear Regression from scratch
- Linear Regression, Ridge and Lasso with sklearn
- Compare models

Mini Project: House Price Prediction.

Metrics: MAE, MSE, RMSE and R^2 .

DAY 6 – Classification Fundamentals**Algorithms**

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Naive Bayes

Mathematics

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Odds and log-odds
- Decision boundary
- Log loss / cross-entropy

Coding

- Train all three models on the same dataset
- Compare predictions and evaluation metrics

Suggested Dataset: Breast Cancer dataset.

Daily Output: Classification comparison table + working notebook.

DAY 7 – Decision Trees + Random Forest**Decision Trees**

- Nodes, splits and leaves
- Entropy
- Information gain
- Gini impurity
- Tree depth and pruning

$$H(S) = - \sum_i p_i \log_2 p_i \quad G = 1 - \sum_i p_i^2$$

Random Forest

- Bagging
- Bootstrap samples
- Random feature selection
- Voting and averaging

Coding: Decision Tree + Random Forest; tune `max_depth` and `n_estimators`.

Interview: Why can Random Forest reduce overfitting compared with a single deep tree?

DAY 8 – SVM + Kernel Methods**Topics**

- Hyperplanes
- Margins
- Support vectors
- Hard vs soft margin
- C parameter
- Hinge loss
- Kernel trick

Kernels

- Linear
- Polynomial
- RBF

Mathematics

- Inner product
- Distance from a point to a hyperplane
- Optimization intuition

Coding: Compare Logistic Regression, SVM and Random Forest on one dataset.

Daily Output: SVM intuition diagram + comparison notebook.

DAY 9 – Ensemble Learning**Bagging**

- Bootstrap sampling
- Independent base learners
- Aggregation

Boosting

- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

Key ideas

- Weak learners
- Learning rate
- Number of estimators
- Tree depth
- Subsampling

Core distinction: Bagging trains models independently; boosting builds models sequentially to correct previous errors.

Coding: Train a gradient boosting/XGBoost model and compare it with Random Forest.

Daily Output: Ensemble learning comparison chart.

DAY 10 – Model Evaluation + Cross Validation**Classification Metrics**

- Confusion matrix
- Accuracy
- Precision
- Recall
- F1 score
- Specificity
- ROC-AUC
- PR-AUC
- Log loss

Regression Metrics

- MAE, MSE, RMSE
- R^2
- Adjusted R^2
- MAPE

Cross Validation

- K-Fold
- Stratified K-Fold
- Leave-One-Out
- Time Series Split

Critical Concept: Data leakage and how to prevent it.

Daily Output: Metric decision guide: “Which metric should I use and why?”

DAY 11 – Clustering**Topics**

- K-Means
- Centroids
- Inertia
- Elbow method
- Silhouette score
- Hierarchical clustering
- DBSCAN
- Gaussian Mixture Models

Understand: K-Means vs DBSCAN and when each is appropriate.

Coding

- Scale customer data
- Apply K-Means
- Select K using elbow/silhouette analysis
- Analyze cluster characteristics

Mini Project: Customer Segmentation.

Daily Output: Cluster visualization + interpretation.

DAY 12 – PCA + Dimensionality Reduction**Mathematics**

- Covariance matrix
- Eigenvalues
- Eigenvectors
- Variance
- Orthogonality

PCA Pipeline

High Dimensions $\rightarrow$ Covariance $\rightarrow$ Eigenvectors $\rightarrow$ Principal Components $\rightarrow$ Reduced Dimensions

Also Study

- t-SNE
- UMAP (conceptual understanding)

Coding: Reduce a high-dimensional dataset to 2D with PCA and visualize it.

Daily Output: PCA explanation with eigenvalue/eigenvector intuition.

DAY 13 – Feature Engineering + Feature Selection**Feature Engineering**

- Numerical transformations
- Log transformation
- Polynomial features
- Binning
- Interaction features
- Date/time features
- Text features

Feature Selection

- Correlation
- Variance threshold
- Mutual information
- Chi-square
- Recursive Feature Elimination (RFE)
- L1 regularization

Advanced Concept: Curse of dimensionality.

Coding: Improve one dataset by engineering and selecting better features.

Daily Output: Before/after model comparison.

DAY 14 – Hyperparameter Optimization

Understand

- Parameters vs hyperparameters
- Why validation data is required
- Avoiding test-set leakage

Techniques

- Grid Search
- Random Search
- Bayesian Optimization
- Optuna (practical/conceptual)

Examples of Hyperparameters

- Learning rate
- Tree depth
- Number of estimators
- Regularization strength

Coding: Use `GridSearchCV` and `RandomizedSearchCV`.

Daily Output: Tuned model + documented best parameters.

DAY 15 – Imbalanced Learning

Topics

- Class imbalance
- Undersampling
- Oversampling
- SMOTE
- Class weights
- Threshold tuning

Key Idea: Accuracy can be misleading when one class is rare.

Focus Metrics

- Precision
- Recall
- F1
- PR-AUC

Mini Project: Credit Card Fraud Detection.

Coding: Compare class weighting, resampling and threshold changes.

Daily Output: Imbalanced-classification experiment report.

DAY 16 – Anomaly Detection + Advanced ML**Anomaly Detection**

- Isolation Forest
- One-Class SVM
- Local Outlier Factor (LOF)

Semi-Supervised Learning

- Labelled vs unlabelled data
- Pseudo-labeling

Calibration

- Calibration curves
- Brier score
- Platt scaling
- Isotonic regression

Coding: Detect anomalies in a synthetic or real dataset and visualize the results.

Daily Output: Anomaly detection comparison + calibration notes.

DAY 17 – Time Series Machine Learning**Topics**

- Trend
- Seasonality
- Noise
- Lag features
- Rolling statistics
- Time-based splitting

Traditional Methods

- ARIMA – conceptual
- SARIMA – conceptual

ML Methods

- Random Forest
- Gradient Boosting
- XGBoost

Critical Rule: Do not randomly shuffle ordinary forecasting data. Train on the past and validate/test on the future.

Mini Project: Sales Forecasting.

Daily Output: Time-aware train/validation pipeline.

DAY 18 – NLP + Recommendation Systems**Classical NLP**

- Bag of Words
- TF-IDF
- N-grams
- Text preprocessing
- Naive Bayes for text
- Logistic Regression for text
- Linear SVM for text

Recommendation Systems

- Content-based filtering
- Collaborative filtering
- User-item matrix
- Cosine similarity
- Matrix factorization
- Cold-start problem

Mini Project: Movie Recommendation System.

Coding: Build a content-based recommender using TF-IDF/cosine similarity and understand collaborative filtering conceptually.

Daily Output: Working recommendation prototype.

DAY 19 – Explainable ML + MLOps**Explainability**

- Feature importance
- Permutation importance
- Partial dependence
- SHAP
- LIME

ML Engineering

- Model serialization
- Pipelines
- Experiment tracking
- Model versioning
- Data versioning
- Data drift
- Model drift
- Monitoring and retraining

Deployment Concepts

Data → Preprocessing → Model → API → Docker → Cloud → Monitoring

Coding: Save a trained model, expose a prediction function/API conceptually, and document the monitoring plan.

Daily Output: One-page ML lifecycle diagram.

DAY 20 – Master ML Revision + Capstone**Part 1 – Mathematics Test**

- Vectors, matrices, dot product, norms
- Eigenvalues/eigenvectors
- Covariance matrix
- Derivatives, partial derivatives and gradients
- Chain rule and gradient descent
- Conditional probability and Bayes theorem
- Expected value, variance and covariance
- Distributions, standard deviation and Z-score

Part 2 – Algorithm Recall

Explain without notes:

- Linear Regression
- Logistic Regression
- KNN
- Naive Bayes
- Decision Tree
- Random Forest
- SVM
- K-Means
- DBSCAN
- PCA
- Gradient Boosting
- XGBoost
- Isolation Forest

For every algorithm answer:

1. What is it?
2. How does it work?
3. What is the mathematical intuition?
4. When should it be used?
5. Advantages and disadvantages?
6. Important hyperparameters?
7. Which metrics should evaluate it?

Part 3 – 3-Hour Capstone

Build an end-to-end project:

Dataset → EDA → Cleaning → Feature Engineering → Split → Preprocessing → Baseline → Advanced Model →

Final Test: Explain your entire project aloud as if you are in an ML interview.

Daily Output: Complete capstone + final ML cheat sheet + list of weak topics for future revision.

20-Day Project Portfolio

Day	Project / Practical Output
2	Linear Regression + Gradient Descent from Scratch
5	House Price Prediction
6	Classification Model Comparison
9	Ensemble Learning Comparison
11	Customer Segmentation
12	PCA Visualization
15	Credit Card Fraud Detection
17	Sales Forecasting
18	Movie Recommendation System
20	End-to-End ML Capstone

Master Checklist

Mathematics

- ☐ Linear algebra
- ☐ Vectors and matrices
- ☐ Dot products and norms
- ☐ Eigenvalues/eigenvectors
- ☐ Covariance
- ☐ Calculus
- ☐ Derivatives
- ☐ Gradients
- ☐ Chain rule
- ☐ Optimization
- ☐ Probability
- ☐ Bayes theorem
- ☐ Statistics
- ☐ Distributions

Core ML

- ☐ Linear Regression
- ☐ Polynomial Regression
- ☐ Ridge
- ☐ Lasso
- ☐ Logistic Regression
- ☐ KNN
- ☐ Naive Bayes

- ☐ Decision Trees
- ☐ Random Forest
- ☐ SVM

Advanced ML

- ☐ Bagging
- ☐ AdaBoost
- ☐ Gradient Boosting
- ☐ XGBoost
- ☐ LightGBM
- ☐ CatBoost
- ☐ K-Means
- ☐ Hierarchical clustering
- ☐ DBSCAN
- ☐ GMM
- ☐ PCA
- ☐ t-SNE
- ☐ UMAP
- ☐ Isolation Forest
- ☐ One-Class SVM
- ☐ LOF

ML Engineering

- ☐ Pipelines
- ☐ Cross-validation

- ☐ Hyperparameter tuning
 - ☐ Grid Search
 - ☐ Random Search
 - ☐ Bayesian optimization
 - ☐ Feature engineering
 - ☐ Feature selection
 - ☐ SMOTE
 - ☐ Calibration
 - ☐ SHAP
 - ☐ LIME
 - ☐ Data drift
 - ☐ Model drift
 - ☐ Deployment
 - ☐ Monitoring
- Specialized ML**
- ☐ NLP
 - ☐ TF-IDF
 - ☐ Recommendation systems
 - ☐ Time series
 - ☐ Anomaly detection

Final Goal

By the end of Day 20, you should be able to:

- Explain the mathematics behind major ML algorithms.
- Implement important algorithms using Python and scikit-learn.
- Build complete preprocessing and modeling pipelines.
- Select appropriate evaluation metrics.
- Detect overfitting, leakage and data-quality problems.
- Tune models systematically.
- Explain advanced ML techniques in interviews.
- Build and present an end-to-end ML project.

Learn → Recall → Code → Explain → Apply

20 Days. One Complete ML Revision Cycle.